04/08/2022

Friday, April 8, 2022 11:29 AM

M168_Notes_8.3

8.3 Variance and Standard Deviation

3 3 3 3 3 2 2 3 4 4 0 0 5 5 5

X	$X-\mu$	$(X-\mu)^2$	
2	-1	1	
2	-1	1	
3	0	0	
4	1	1	
4	1	1	

$E(X) = \mu \leftarrow$ Mean

$\mu = 3 = E(X)$

$5 = n =$ number of datapoints/samples.

POPULATION VARIANCE,

$$Var(X) = \sum \frac{(X-\mu)^2}{n} = \frac{4}{5}$$

POPULATION STANDARD DEVIATION,

$$\sigma(X) \qquad \sigma = \sqrt{Var(X)} = \sqrt{\frac{4}{5}} \approx 0.894$$

What do we mean by population?

This means everyone, so if ALL the members of the population are used to find the mean, we use the symbols μ and σ.

μ ← Population mean

The mean from SAMPLE uses the symbol $\bar{x}$ ← Sample mean.

SAMPLE STANDARD DEVIATION,

$$S = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n-1}}$$ Sample standard deviation.

When we have the probability, it is assumed we had the entire population to base it on, so it is appropriate to use μ and σ.

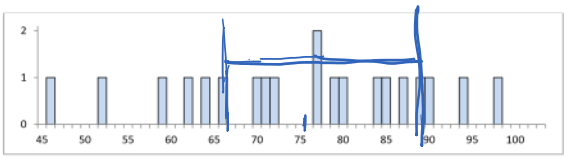
Example

Find the mean and standard deviation for the following distribution:

X	P(X)
−2	0.2
−1	0.3
0	0.1
1	0.1
2	0.3

Example

An exam has an mean of μ=75 and a population standard deviation of σ=14.

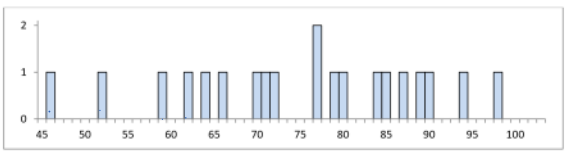


What is the probability that a randomly chosen data point is within 1 standard deviation of the mean?

$75 - 14 = 61$, $75 + 14 = 89$

$14/20 \approx 70\%$ 0.7

What is the probability that a randomly chosen data point is within 2 standard deviations of the mean?

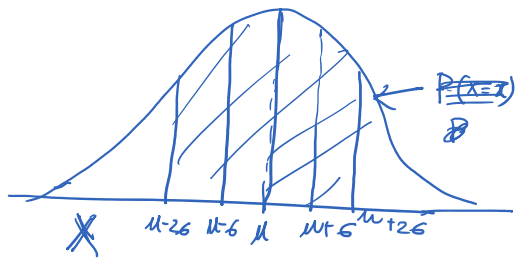


$75 - 2 \times 14$ $75 + 2 \times 14$

47 103 $\frac{19}{20} \approx 95\%$ $\boxed{0.95}$

Chebychev's Theorem: For any data distribution with mean μ and standard deviation σ, the probability that a randomly chosen data point is within $\mu - \kappa\sigma$ to $\mu+\kappa\sigma$ is at least $1 - 1/\kappa^2$. Or,

$$P(\mu-\kappa\sigma \le x \le \mu+\kappa\sigma) \ge 1 - {}^{1}/_{\kappa^2}$$



Example

A probability distribution has a mean of 50 and a standard deviation of 5.

$\mu = 50$

$\sigma = 5$

a) What is the probability that an outcome of the experiment lies between 35 and 65?

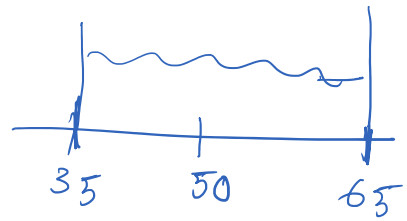
35 65

$$P \ge 1 - \frac{1}{9} = \frac{8}{9}$$

$\mu - k\sigma = 35$

$50 - k \cdot 5 = 35$

$\Rightarrow k = 3$



b) Find the value of k so that at least 93.75% of the data lies in the range 50-5k to 50+5k.

$$1 - \frac{1}{k^2} \ge \frac{93.75}{100}$$

$$1 - \frac{93.75}{100} \ge \frac{1}{k^2} \Rightarrow \frac{6.25}{100} \ge \frac{1}{k^2} \Rightarrow \frac{2.5}{10} \ge \frac{1}{k}$$

$$1 - \frac{93.75}{100} = \frac{1}{k^2} \longrightarrow k = 4.$$

$$\Rightarrow 0.25 \ge \frac{1}{k} \Rightarrow \boxed{k \ge 4}$$

Great!